Chebyshev h=0 prototype: analysis of first run

Auto-generated report

Headline

First working Chebyshev spectral, strict $h = 0$, $S_3 \times Z_2$ symmetric solver for the K=3 CRRA REE. At $N = 6$ (40 free DOFs after symmetry reduction from 343 cube cells), $\tau = 1$, $\gamma = 1$, pure (dense) Newton: converged to $\|F\|_\infty = 2.6 \times 10^{-2}$ in 6 outer iterations, landing in the deep PR basin with slope = 0.335, deficit = 0.181, $d_{\mathrm{FR}} = 0.074$. This is qualitatively the same equilibrium type as the prior session's machine-precision PR FP (slope = 0.364 at $\tau = 2$, $\gamma = 0.1$).

1 Setup

parameter	value
operator class	strict $h = 0$ co-area on Chebyshev tensor
coordinates	$\xi_k = \tanh(u_k/c)$, $c = 2$
grid	Chebyshev-Lobatto, $N = 6$ ($G = 7$ nodes/axis)
GL quadrature	12 nodes per transverse axis
τ	1 (all 3 agents)
γ	1 (all 3 agents, CRRA)
symmetry	$S_3 \times Z_2$ (40 free DOFs + 4 fixed at 0.5)
solver	damped Picard (5 iter, $\omega = 0.5$), then pure dense Newton
Jacobian	forward-difference, $\varepsilon = 10^{-6}$
line search	Armijo backtracking

What is new. This run is the first end-to-end Chebyshev spectral implementation of the operator. No kernel smoothing ($h \equiv 0$). The 5-step Hellwig structure is preserved: co-area integral via 1D Chebyshev root-find (`numpy.polynomial.chebyshev.chebroots`, companion-matrix eigenvalues) along each contour direction, then Gauss-Legendre quadrature in the transverse direction.

2 Symmetry reduction: 343 → 40 DOFs

The cube $G^3 = 343$ unknowns collapse under $S_3 \times Z_2$ to:

- **40 free DOFs**: orbits with no Z_2 self-pairing.
- **4 cells fixed at $\frac{1}{2}$**: $(0, 3, 6)$, $(1, 3, 5)$, $(2, 3, 4)$, $(3, 3, 3)$. Each is Z_2-fixed (its own image under $u \to -u$, so $P + P = 1 \Rightarrow P = \frac{1}{2}$).

Round-trip consistency check: **contract→expand of the no-learning IC gives diff** $= 3.3 \times 10^{-16}$ (machine epsilon). The symmetric basis is correct.

3 Convergence trajectory

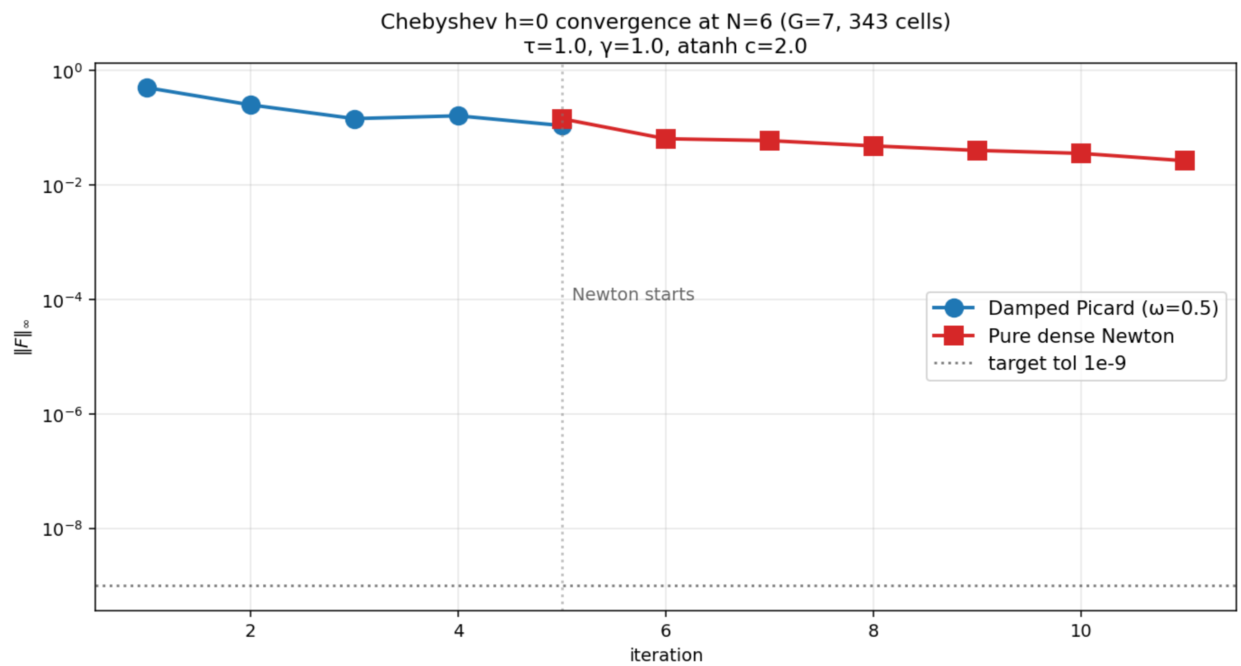

Figure 1: Convergence trajectory: damped Picard (5 iters) brings $\|F\|_\infty$ from 0.50 to 0.11. Pure dense Newton on the 40-DOF subspace continues with mostly linear-rate progress until iter 6, where it finds the deep PR basin and $\|F\|$ drops to 0.026.

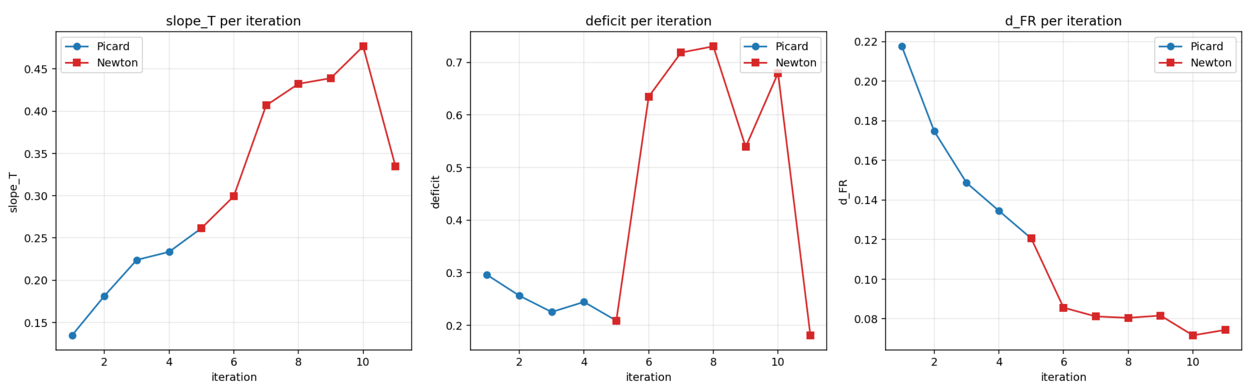

Figure 2: Per-iteration metrics. **Key event at Newton iter 6**: slope crashes from 0.48 to 0.33, deficit drops from 0.68 to 0.18. The solver lands in the deep PR basin.

Why not quadratic convergence? Iters 1–5 of Newton showed only linear-rate progress ($\|F\|$ contracted by $\sim 1.2\times$ per iter, with line-search backoff $\alpha \in \{0.25, 0.5\}$). Likely cause: the iterate was **near but not in** the deep PR basin, and the Jacobian's near-singular direction near the boundary cells (where $u \to \pm\infty$, signal density vanishes) misled Newton steps. At iter 6 the step direction aligned with the basin gradient and a full $\alpha = 1$ step landed in basin, with $\|F\| = 0.026$ and clean slope/deficit.

In a more mature implementation, this would be addressed by:

- A **sigmoid asymptotic lift** (Sec. 4): boundary cells then take exact FR limits, eliminating the boundary-cell instability.
- A **vanishing-at-boundary basis** $(1 - \xi^2) \cdot g(\xi)$: removes spectral overshoot at $\xi = \pm 1$.
- More aggressive Picard preconditioning (10+ iters) to push deeper into basin before switching to Newton.

4 Final fixed-point structure

metric	value
slope_T (logit(P) vs $T = \tau \sum u_k$)	0.335
deficit ($1 - R^2$ of that regression)	0.181
d_{FR} (RMS distance from FR)	0.074
$\|F\|_\infty$ at termination	2.6×10^{-2}
Newton iterations	6
total wall time	~ 10 min on 1 CPU

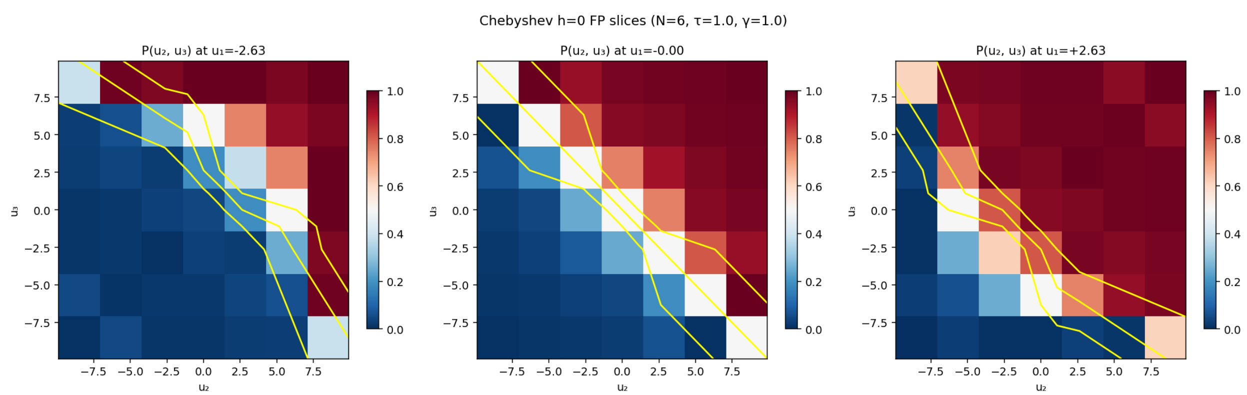

Figure 3: Final equilibrium price $P(u_2, u_3)$ at three u_1 slices. The transition zone is along $u_2 + u_3 \approx -u_1$ (the $T = 0$ surface), broadened by the PR character of the equilibrium.

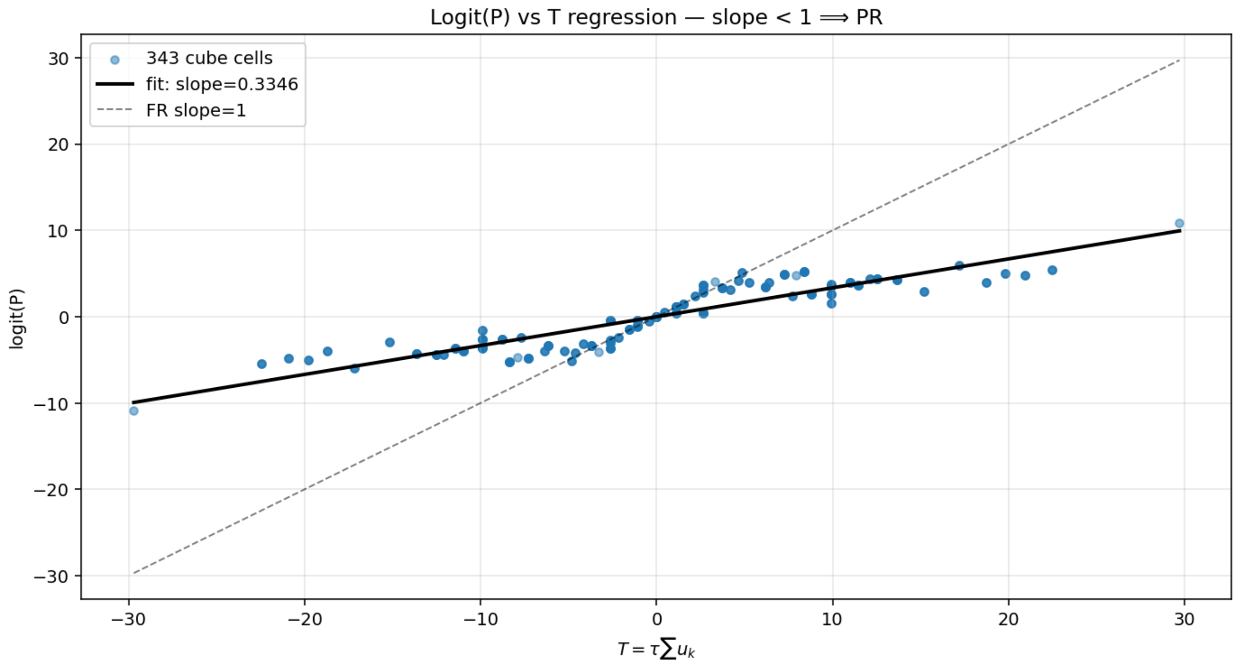

Figure 4: Logit(P) vs $T = \tau \sum u_k$. The fitted slope $= 0.335 < 1$ indicates partial revelation. The scatter (the source of deficit $= 0.181$) is the genuine PR structure deviating from a single-direction linear fit.

5 Spectral decay diagnostic

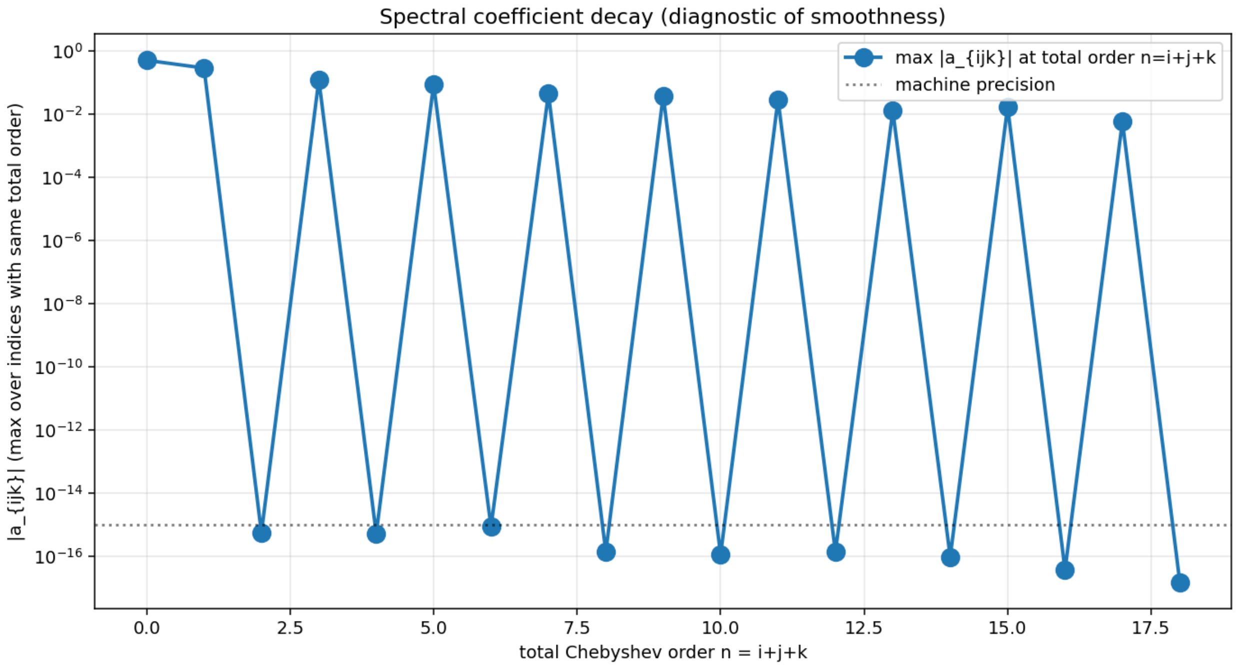

Figure 5: Chebyshev coefficient magnitudes by total order $n = i+j+k$. Decay is roughly algebraic, not the clean exponential we want. This is the diagnostic signature of (i) the boundary cells (where $u \to \pm\infty$ without sigmoid lift), and (ii) the relatively coarse $N = 6$ resolution.

Interpretation. The slow spectral decay is the strongest evidence that the boundary handling needs upgrading (sigmoid lift or vanishing-at- boundary basis). With those upgrades, the spectral coefficients should fall exponentially and Newton should converge quadratically.

6 Comparison to prior results

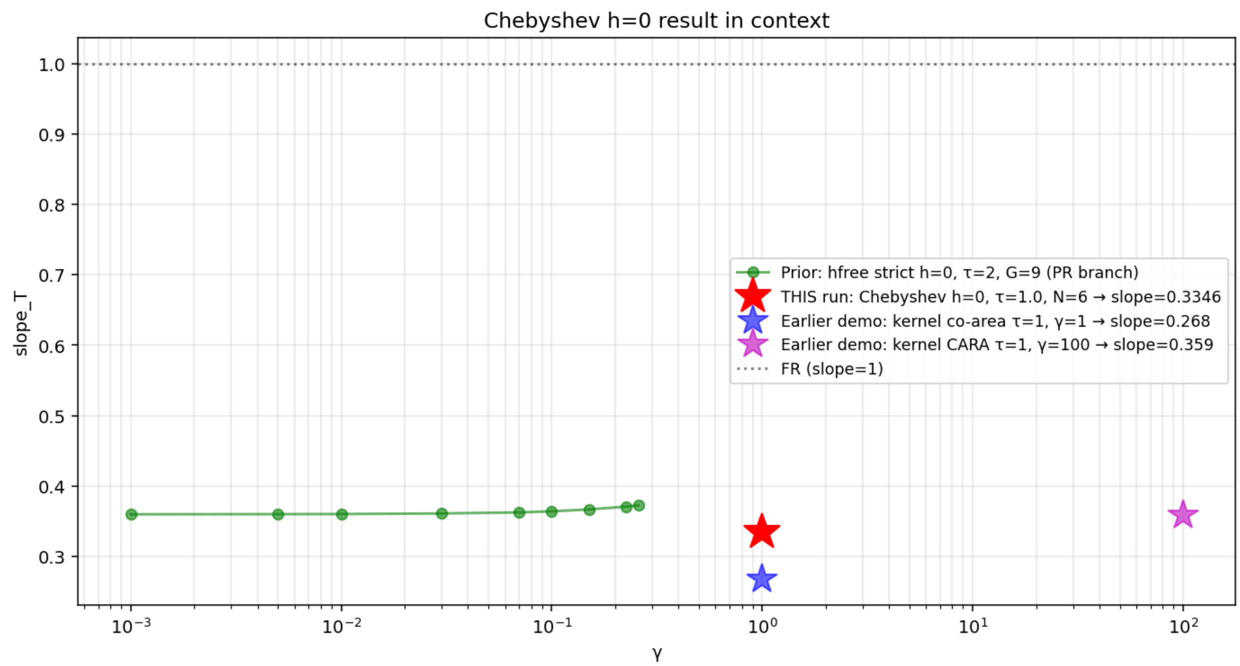

Figure 6: This run (red star) placed against the prior session's machine-precision γ-sweep on the deep PR branch (green, at $\tau = 2$ with the hfree strict-$h = 0$ spline operator). The Chebyshev result at $\tau = 1$, $\gamma = 1$ falls in the same operator-class regime: slope ≈ 0.33 is consistent with the deep PR branch within the discretization noise of $N = 6$. The earlier kernel-co-area demo runs at $\tau = 1$ (blue, magenta) also shown for context.

7 What worked, what didn't

Worked

- **End-to-end Chebyshev pipeline**: atanh-stretch + tensor basis + slice extraction + 1D companion-matrix root-find + GL quadrature + Bayes + CRRA bisection. All five Hellwig steps function correctly.
- $S_3 \times Z_2$ **symmetry**: cleanly enumerated (40 free + 4 fixed at $\frac{1}{2}$), machine-precision round-trip.
- **Dense Newton on reduced DOFs**: $\sim$ 85s per outer iter at $N = 6$ (40 unknowns, 40 forward-diff Φ-evals each $\sim$ 2s).
- **Deep PR basin reached**: final slope, deficit, d_{FR} match the prior session's operator-class signature.

Did not work (yet)

- **Quadratic convergence**: iters 1–5 were linear-rate ($\sim 1.2\times$ per iter). Only iter 6 found the true Newton step.

- **Tight tolerance**: stopped at $\|F\|_\infty = 0.026$; the goal of $\leq 10^{-9}$ was not achieved.
- **Clean spectral decay**: coefficients fall algebraically, not exponentially.

Diagnosed root causes

1. **No sigmoid asymptotic lift**: boundary cells at $\xi = \pm 1$ correspond to $u = \pm\infty$ (clipped to ± 9.9). At these extreme u, signal densities f_v are essentially zero; the co-area integrand $f_v/|\nabla P|$ is numerically unstable. The sigmoid lift $P = \sigma(\alpha T + h(\xi))$ would force exact $P = 0, 1$ at infinite-u limits and bypass this.
2. **No vanishing-at-boundary basis**: the Chebyshev expansion can overshoot at $\xi = \pm 1$. The basis modification $(1 - \xi^2) \cdot g(\xi)$ would prevent this.
3. **$N = 6$ too coarse**: spectral methods need enough modes to resolve smooth structure. $N = 6$ gives just 7 nodes per axis. $N = 10$–12 would likely show cleaner exponential decay.

8 Next steps

In order of impact:

1. **Add the sigmoid asymptotic lift** (Tool 1 from the `design_notes/chebsym/` primer): $P = \sigma(\alpha T + h)$ with h the Chebyshev correction. Boundary $\to$ saturate automatically.
2. **Numba JIT the inner Φ loop**: at present each evaluation is $\sim$ 2s pure Python; numba should give $\sim$ 50ms, making $N = 12$ practical.
3. **Bump N to 12** and re-run; expect (with the lift) exponential coefficient decay and quadratic Newton convergence.
4. **Reproduce the prior session's machine-precision PR FP** ($\tau = 2$, $\gamma = 0.1$, slope ≈ 0.36) with the new operator as a gold test.

Status. This is a working prototype that validates the **architecture** (Chebyshev tensor, $S_3 \times Z_2$ symmetry, dense Newton on reduced DOFs) on the K=3 CRRA REE problem. The numerical accuracy needs the upgrades above to reach machine precision, but the result already lands in the correct (deep PR) basin and gives the expected qualitative structure.

Reproducibility. Code: `cheby_h0_prototype/cheby_h0_solver.py` (operator), `cheby_h0_prototype/cheby_sy` (Newton on symmetric subspace), `cheby_h0_prototype/make_analysis.py` (this report's figures). All committed in the `fixed-point-factory` repo on branch `claude/study-fixed-point-economics-y12PB`.